Theoretical and Applied Climatology (2022) 149:695–708
https://doi.org/10.1007/s00704-022-04073-w

ORIGINAL PAPER

Optimal selection of representative climate models and statistical downscaling for climate change impact studies: a case study of Rhode Island, USA

Shiva Gopal Shrestha[1] · **Soni M. Pradhanang**[1]

Received: 24 March 2021 / Accepted: 20 April 2022 / Published online: 7 May 2022
© The Author(s), under exclusive licence to Springer-Verlag GmbH Austria, part of Springer Nature 2022

Abstract
The future climate impact studies rely on future projections obtained from downscaling of Coupled Model Intercomparison Project Phase 5 (CMIP5) climate models. The main challenge is large, yet growing pool of CMIP5 models, posing a high computational cost for analyzing the climate model ensemble. The optimal model selection from the large pool is hence necessary. It is critical that the selected climate models represent the uncertainty range of possible future and have high skill in past performance. In this study, we use a multi-criteria decision process based on the coupling of the envelope and past performance approach to identify potential future climate stresses for RI, USA. The selection is based on projected changes in annual climatic means (precipitation and temperature) followed by a range of projected changes in climatic extremes and past performance among these models. From a pool of 109 models from RCP4.5 and 79 models from RCP8.5, a final subset of 4 models was selected for RCP4.5 and RCP8.5 respectively. The change in annual climatic means for temperature varied + 1.7 to + 3.0 °C in RCP4.5 and + 2.4 to + 5.8 °C in RCP8.5, and the range in climate mean of annual precipitation varied from 0.2 to 12.7% in RCP4.5 and − 4.1 to 15.9% in RCP8.5. The selected climate models are statistically downscaled to produce reliable local-scale climate estimates. Various variants of quantile mapping are studied, and quantile delta mapping is applied to systematically reduce biases and preserve raw GCM signals.

1 Introduction

A deeper understanding and analysis of current and future climate at local, regional, and global scales is central to understanding the possible impacts on humans and the environment. The spatial variability in topography, land use, and different human interventions like agriculture induces a different response to different climatic forcing (Hayhoe et al. 2006). The analysis of US northeast using the United States Historical Climatology Network sites has shown significant changes in temperature and precipitation and the hydrological component of the water cycle such as snowfall, snow to rain ratio, snowmelt from 1948 through 2000 (Hodgkins and Dudley 2006a, 2006b; Hodgkins et al. 2003), and, in turn, water quantity and water quality variables of lakes and rivers (Hodgkins et al. 2002, 2003; Tu 2009). Anthropogenic factors are extremely likely to trigger climate change, and climate change triggered by anthropogenic factors is projected to continue severely impacting humans and the environment (Change 2013). Climate impact studies under these forcing help generate future scenarios that policymakers may use to develop adaptation measures accordingly (Rosen and Guenther 2015).

Global climate models (GCMs) are developed under Coupled Model Intercomparison Project (CMIP) to understand physical processes and simulate physical hydrological processes at a global scale. These models can generate climate change scenarios of past and future time slices (Eden et al. 2014; Hawkins and Sutton 2009; Taylor et al. 2012). CMIP5 archive (Taylor et al. 2012) indicates that there is a large model matrix of 61 different GCMs, each indicating a potential future climate forcing. Due to increasing interest in climate science, the ensemble pool of GCMs is growing in number and model complexity, and understanding has been evolving such that relevant processes are being incorporated (Flato et al. 2014; Gulizia and Camilloni 2015; Knutti et al.

✉ Soni M. Pradhanang
spradhanang@uri.edu

Shiva Gopal Shrestha
Shiva_shrestha@uri.edu

[1] Department of Geosciences, University of Rhode Island, 9 E. Alumni Ave, Kingston, RI 02881, USA

2013). Despite the robustness and increased confidence in the climate models, the inherent uncertainty in these models is still large (Friedlingstein et al. 2014; Knutti and Sedláček 2012).

The gradually increasing climate models and their multiple ensemble members have introduced constraints in climate impact studies due to the high demand for computational and human resources. The general trend has been to work with a small ensemble of climate models or a single model or multi-model mean. However, these approaches do not truly represent the variability and uncertainty of the model pool. This has introduced a big hurdle in climate impact studies, and a selection of representative ensemble models that can represent the full ensemble of climate models has been challenging.

2 Study area

The study was performed to support the objective of simulating hydrological models under potential future climate scenarios in watersheds of RI, USA (Fig. 1). The fastest warming hotspot in the USA is the northeast, which is projected to temperature increase by 3 °C when global warming reaches 2 °C (Karmalkar and Horton 2021). Rhode Island has experienced the highest change in temperature in this region (Kunkel et al. 2022). Rhode Island is the state in eastern USA having a dimension of 60 km breadth and 77 km length. GCMs depict the climate using a grid over the globe having a horizontal resolution of between 250 and 600 km. Even a single grid cell of the GCM would cover the study area unless the boundary passes through the study area which has been considered during this study. This would encourage for GCM selection in Rhode Island rather than the northeastern USA.

3 Methodology

3.1 Multi-criteria model selection

The two widely used approaches for climate model selection are the skill-based approach and envelop approach. A model may be considered skilled in skill-based approach if it can replicate past and present observed climatic characteristics and underlying processes (Flato et al. 2014; Hidalgo and Alfaro 2015; Pierce et al. 2009). The model's ranking based on types and number of metrics may yield different selections (Brekke et al. 2008; Hidalgo and Alfaro 2012). Envelope approaches intend to preserve the variability in a wide range of available projections. These approaches selected the climate variables under consideration and aimed at maintaining the characteristics of distributions of the full ensemble(Ruane and McDermid 2017). These approaches are robust as it preserves the range of plausible outcomes and model-based uncertainty at the same time. A similar study corroborates choosing optimal subset ensembles that

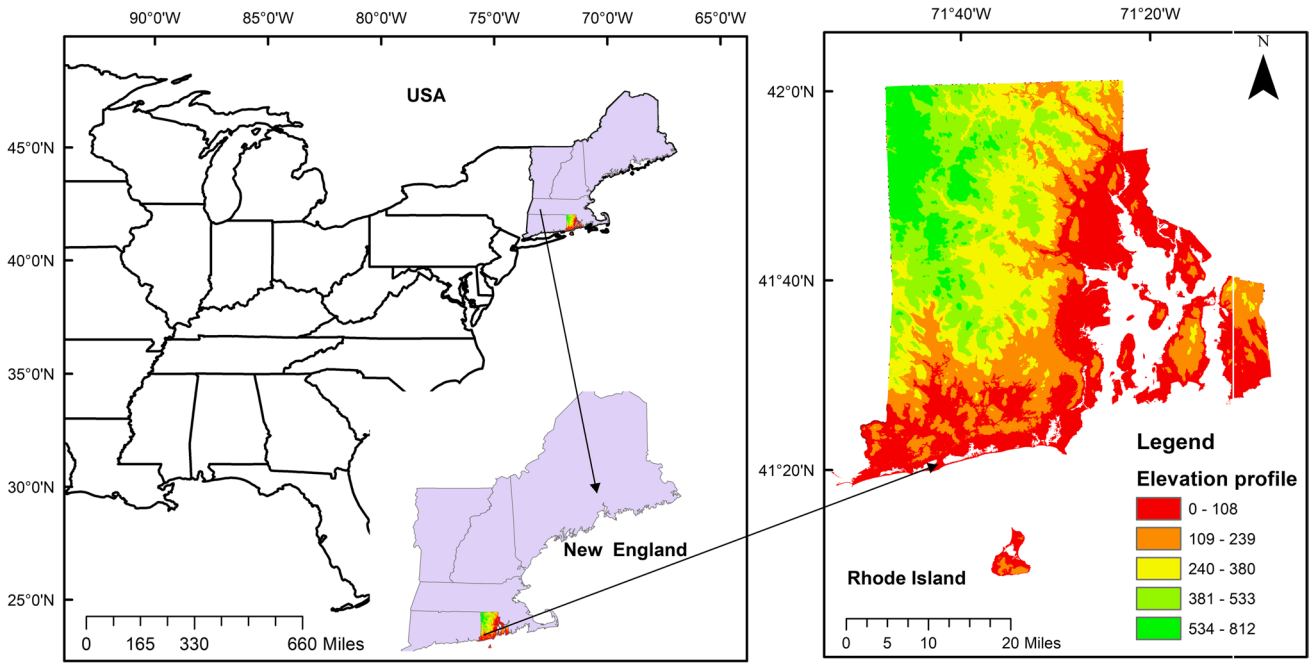


Fig. 1 Study area as Rhode Island in New England region

would retain the full ensemble's mean and variability with minimum bias (Evans et al. 2013).

Envelope-based approaches successfully preserve variability in climate models' projections but do not consider model skill. Similarly, skill-based approaches are successful in choosing the model that is better in past performance but do not consider the model variability. When subsetting GCMs from the large ensemble, important information may be lost, but these approaches may yield optimal subset under specific identification criteria and, therefore, capture adequate variability of the model pool (Liu et al. 2013; McSweeney and Jones 2016). For better efficacy and representation, multiple criteria model selection need to be pursued considering the objective of the study (Lutz et al. 2016). The choice of the climate variables will also vary accordingly. Some studies incorporate the mean air temperature and annual precipitation as major climatic variables (Immerzeel et al. 2013; Sorg et al. 2014; Warszawski et al. 2014). Here, we propose a novel approach that would employ both envelope and model skill-based criteria. The aim of the study is to address the high degree of uncertainty inherent in the pool of climate models. For this reason, we may not be able to come with the most skillful solution, but we want our solution or the optimal model matrix lies on the areas which is more plausible and addresses the concerns of uncertainty. In order to overcome the model skill concern, we employ bias correction techniques to bridge this gap. This will help select an optimal subset of a model representing all possible futures through climatic means and extremes in precipitation and temperature. The chosen model subset will be efficient and practical in climate impact studies and simultaneously portray the potential future climate scenarios.

The model selection method discussed here coupled the envelope approach and model skill approach implemented in three steps (Fig. 2). The first step of model selection was based on the criteria of average annual changes in the mean annual precipitation and temperature to capture the climate regime. Likewise, the second step focuses on the model selection was based on changes in climatic extremes indices to capture extreme weather regime. The last step was selected a better-skilled model that could properly simulate the past and present climate scenarios.

3.1.1 Initial selection (step 1): selection based on changes in climatic means

In AR5 2014, IPCC introduced a greenhouse gas concentration trajectory known as the representative concentration pathway (RCP) for representing long-term and near-term climate modeling experiments (Change 2013). The RCP scenarios consider four emission scenarios, which are (a) one mitigation scenario (RCP2.6), (b) one very high baseline emission scenario (RCP8.5), and (c) two medium stabilization scenarios (RCP4.5/RCP6.0) (Van Vuuren et al. 2011).

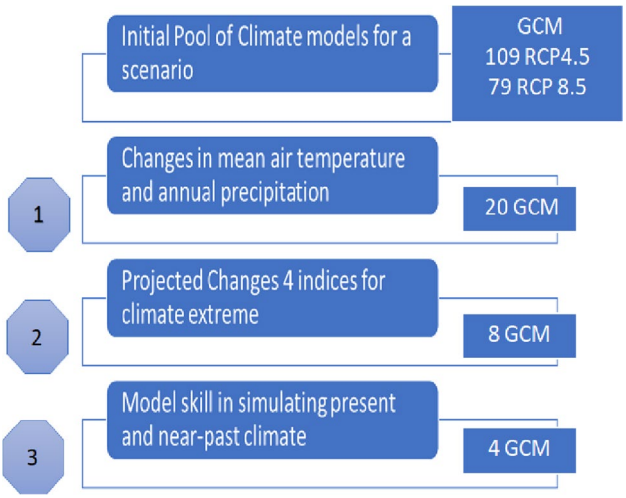


Fig. 2 Flow diagram showing model selection procedure

The Paris Agreement is an international treaty on climate change which was adopted by the USA in 2016.The objective of this agreement is to limit global warming to below 2, preferably to 1.5 °C, compared to pre-industrial levels. As per this agreement, the pathway is RCP1.9 which is a pathway that limits global warming to below 1.5 °C.

Many studies in climate change have chosen a medium stabilization scenario as RCP4.5 and high baseline emission scenario as RCP8.5 as plausible range of future climate (Lutz et al. 2016; San José et al. 2016). RCP2.6 is a stringent pathway which requires that carbon dioxide (CO_2) emissions start declining by 2020 and go to zero by 2100. RCP4.5 is medium stabilization scenario which represents probable baseline scenario, and RCP8.5 is the most aggressive scenario representing the high emission scenario. We included RCP4.5 and RCP8.5 in this study as these two pathways represented potential future climate scenarios.

The future climate scenarios for our study were selected from the ensemble of climate models based on the RCP4.5, and the RCP8.5 model runs from the CMIP5 repository (Taylor et al. 2012). The model pool consisted of 109 models from RCP4.5 and 79 models from RCP8.5. The model selection process first focused on the change in climate means and extreme variables. The database of projected changes in climatic extreme developed by Sillmann (2013a, b) was used as a reference. The climate extreme indices were used as defined by CCI/CLIVAR/JCOMM Expert Team on Climate Change Detection and Indices (ETCCDI) (Peterson 2005). For the model skill evaluation of GCM runs, the Parameter-Elevation Regressions on Independent Slopes Model (PRISM) was used as a reference dataset (Daly and Bryant 2013).

All the GCMs required resampling. To find the change in climatic means, we resampled all GCM models to 2.5° × 2.5° grid. The average annual changes in temperature (ΔT) and precipitation (ΔP) in GCM matrix were then computed from 1971 to 2000 and 2071 to 2100 using the Royal Netherlands Meteorological Institute (KNMI) Climate Explorer (http://climexp.knmi.nl). From this matrix of change in climatic means, the 10th and 90th percentile values for ΔT and ΔP were used, representing a potential spread of future scenarios from the combination of cold, warm, dry, and wet scenarios of future climate (Immerzeel et al. 2013; Lutz et al. 2016; Sorg et al. 2014). The four potential scenarios enveloping the model pool were (i) dry-cold, (ii) dry-warm, (iii) wet-cold, and (iv) wet-warm future climates, and for each scenario, five different daily climate model nearest to these scenarios based on Euclidean distance were selected (Lutz et al. 2016). For instance, GCM representing a warm, wet climate were those models whose Euclidean distance in terms of ΔT and ΔP values were least to 90th percentile of ΔT and 90th percentile of ΔP. These percentile values were taken to represent the climate scenario to represent the full ensemble and avoid the outliers. During this first step, 20 GCM runs from an initially available pool of models for each RCP scenario were selected. The initial pool of ensemble members contained (denoted with rxixpx) all runs with the ensemble members with different initializations and physics. The process was repeated for each member, as each member had the potential for any initial condition and could end up in a different future.

3.1.2 Subset selection (step 2): selection based on a change in climatic extremes indices

In this step, the model selection was based on average changes in climatic extremes between 2071–2100 and 1971–2000. The characterization of climatic extremes was done using ETCCDI indices for air temperature and precipitation (Peterson 2005) (Table 1). Here, extreme indices for temperature were defined by warm spell duration index (WSDI) and cold spell duration index (CSDI). Similarly, the precipitation due to extremely wet days (R95pTOT) and the number of consecutive dry days were used to define precipitation extremes.

These four indices were first computed for each year in the future period (2071–2100) and reference period (1971–2000) in the study area from the database prepared by Sillmann, and for the indices absent in the database, the same procedures in the study were followed for computing these indices (Sillmann et al. 2013a, b). These indices needed to be averaged for the time period of 30 years, and the difference in the average of these indices represents variation in climatic extremes. For instance, CDD and CSDI were considered for extreme cold and dry scenarios and WSDI and CDD for extreme warm and dry scenarios. Likewise, WSDI and R95pTOT were considered for extremes in a warm and wet scenario. From the selected model in an initial step, the pertinent indices for dry-cold, wet-cold, dry-warm, and wet-warm future climates were selected, and each of the models was ranked based on their scores. For example, in the cold and dry scenario, the model with the maximum increase in CSDI had a rank score of 5, and the model with a maximum increase in CDD had a score of 5. Based on these scores, an average value of these scores would lead to a final rank score, and a subset of two representative models that obtained the highest scores was selected. This would lead to a total of 8 models for each RCP scenarios.

3.1.3 Final selection (step 3): model skill based on past performance

Model skill score is a measure of any model's ability to reproduce the historical climate such that it retains its pattern and variability (Perkins et al. 2007). There are many techniques involved in evaluating the model performance in various temporal scales, including annual or monthly means and standard deviation. Such techniques might be the most common way, but it is not representative of the entire data distribution. To overcome this shortcoming, the assessment skill of a climate model based on its ability to simulate the distribution is necessary. A major advantage of the distribution-based method is that should a model be skilled in simulating the distribution, this will develop confidence in capturing the current extreme events and grow common in the future.

However, a model with a particular skill-based assessment is not without limitations. Some of the instances are in the frequency of rainy days and spells of hot and cold extreme events. For example, if 2-n length samples are drawn from distribution and shuffled, the distribution would not change. Still, the means and correlation of these series could vary a lot in each trial. On the other hand, Anscombe's quartet is another perfect example where four datasets show very near descriptive statistics (mean, standard deviation, and correlation), but have different distributions and appear very different when plotted (Anscombe 1973). Hence, skill scores should account for correlation and distribution across datasets. A novel approach by combining these skills was proposed to bridge this gap. These skill scores were based on the metrics derived based on earlier works aimed at defining past performance (Kjellström et al. 2010; Perkins et al. 2007; Sánchez et al. 2009; Taylor 2001). Here, the selected subset models were further subsetted through a skill test between the historical GCM and historical reference data for monthly average mean air temperature and monthly precipitation sums for the period 1981–2005.

Table 1 Initial model selection

RCP	Projections	CMIP model	ΔP (%)	ΔT (°C)	ΔWSDI (%)	ΔCDD (%)	ΔCSDI (%)	ΔR95 (%)	*T* index	*P* index	Combined score
RCP4.5	Dry Cold	bcc-csm1-1-m_r1i1p1	2.65	1.55	439.64	−1.8	−99.35	30.78	5	1	3
		CSIRO-Mk3-6-0_r3i1p1	3.68	1.72	430.97	6.57	−100	18.61	3	4	3.5
		bcc-csm1-1_r1i1p1	1.95	2.01	627.05	1.81	−99.71	17.12	4	3	3.5
		CSIRO-Mk3-6-0_r8i1p1	4.75	1.79	262.27	1.12	−100	29.37	3	2	2.5
		CMCC-CM_r1i1p1	0.83	2.12	232.49	12.64	−100	25.35	3	5	4
	Dry Warm	CanESM2_r1i1p1	0.2	2.94	2112.56	17.62	−100	20.52	5	5	5
		CanESM2_r3i1p1	0.11	2.76	1712.08	7.43	−98.31	9.99	3	2	2.5
		CanESM2_r5i1p1	2.82	2.96	2068.74	16.77	−100	23.06	4	4	4
		CNRM-CM5_r1i1p1	−0.27	2.34	871.91	5.25	−100	15.16	1	1	1
		CanESM2_r2i1p1	2.87	2.85	1690.28	11.77	−100	23.05	2	3	2.5
RCP4.5	Wet Cold	CCSM4_r1i1p1	6.17	1.97	559.77	2.68	−100	14.36	2	1	1.5
		EC-EARTH_r8i1p1	5.69	1.4	−9.7	−25	−83.6	29.7	4	2	3
		CSIRO-Mk3-6-0_r7i1p1	8.59	2.25	724.41	0.44	−100	46.5	2	4	3
		CSIRO-Mk3-6-0_r9i1p1	9.36	2.28	1633.87	2.88	−96.46	43.14	3	3	3
		GISS-E2-R_r6i1p3	5.19	1.81	651.97	7.08	−83.17	50.85	5	5	5
	Wet Warm	CSIRO-Mk3-6-0_r5i1p1	11.14	2.59	747.44	−6.88	−100	37.01	2	2	2
		CSIRO-Mk3-6-0_r6i1p1	12.28	2.49	549.62	1.07	−98.66	48.08	1	3	2
		CSIRO-Mk3-6-0_r10i1p1	12.7	2.62	810.93	0.72	−89.7	71.84	3	5	4
		CanESM2_r4i1p1	8.34	3.02	1668.63	6.65	−97.83	52.93	5	4	4.5
		CSIRO-Mk3-6-0_r4i1p1	9.39	2.51	905.43	7.21	−100	27.49	4	1	2.5
RCP8.5	Dry Cold	CMCC-CM_r1i1p1	−4.07	2.37	232.49	12.64	−100	25.35	2	4	3
		inmcm4_r1i1p1	4.28	3.62	573.98	2.86	−91.89	-2.13	4	2	3
		bcc-csm1-1_r1i1p1	1.99	3.66	627.05	1.81	−99.71	17.12	3	1	2
		CNRM-CM5_r1i1p1	−0.07	3.77	871.91	5.25	−100	15.16	2	3	2.5
		IPSL-CM5A-MR_r1i1p1	1.76	4.37	−8.065	53.086	−10.53	21.56	5	5	5
	Dry Warm	CanESM2_r3i1p1	1.41	4.96	1712.08	7.43	−98.31	9.99	2	2	2
		CanESM2_r5i1p1	−0.31	4.94	2068.74	16.77	−100	23.06	3	4	3.5
		CanESM2_r1i1p1	−0.9	4.78	2112.56	17.62	−100	20.52	4	5	4.5
		HadGEM2-ES_r4i1p1	4.51	5.01	2121.05	10.92	−100	14.2	5	3	4
		IPSL-CM5A-LR_r1i1p1	1.85	4.76	950.3	−1.27	−100	28.65	1	1	1

Table 1 (continued)

RCP	Projections	CMIP model	ΔP (%)	ΔT (°C)	ΔWSDI (%)	ΔCDD (%)	ΔCSDI (%)	ΔR95 (%)	*T* index	*P* index	Combined score
RCP8.5	Wet	CSIRO-Mk3-6-0_r3i1p1	10.36	3.75	430.97	6.57	− 100	18.61	4	3	3.5
	Cold	CCSM4_r2i1p1	10.89	3.38	965.44	− 4.11	− 97.77	9.27	5	1	3
		CSIRO-Mk3-6-0_r7i1p1	14.26	3.7	724.41	0.44	− 100	46.5	4	5	4.5
		CSIRO-Mk3-6-0_r5i1p1	15.91	3.81	747.44	− 6.88	− 100	37.01	4	4	4
		CCSM4_r1i1p1	9.05	3.62	559.77	2.68	− 100	14.36	4	2	3
	Wet	HadGEM2-CC_r1i1p1	13.76	5.83	4879.8	0.96	− 95.14	45.56	5	3	4
	Warm	HadGEM2-ES_r2i1p1	14.07	6.22	1533.67	8.04	− 100	48.22	2	4	3
		CanESM2_r4i1p1	13.89	4.76	1668.63	6.65	− 97.83	52.93	3	5	4
		MIROC5_r2i1p1	17.27	4.72	808.57	2.29	− 100	13.09	1	1	1
		IPSL-CM5B-LR_r1i1p1	10.33	5.02	1674.39	0.18	− 87.95	38.5	4	2	3

To address the correlation, the metric for both precipitation and temperature is defined using Taylor's score presented in Eq. (1):

$$T_{\text{taylor}} = \frac{4(1+R)^n}{\left(\frac{\sigma_m}{\sigma_o} + \frac{\sigma_o}{\sigma_m}\right)^2 (1+R_0)^n} \tag{1}$$

where R is a correlation between model and observed data; σ_o and σ_m are the standard deviation of the observations and model, respectively; R_0 is the maximum correlation attainable; and n is the penalty term imposed for low correlation. The higher values of Taylor's score for any model represent a better pattern representation of the observed data.

Due to the different distributional nature of precipitation and temperature, Perkins' and Sanchez's skill scores were required for temperature and precipitation. The final skill score for temperature is an equally weighted score from the Perkins', and Taylor's approach was used (Perkins et al. 2007; Taylor 2001). In the Perkins test, the metric aims to compute the overlap between two PDFs by calculating the cumulative minimum value of two distributions of each binned value (Eq. (2)):

$$T_{\text{perkin}} = \sum_1^n \text{minimum}\left(Z_{\text{MOD}}, Z_{\text{OBS}}\right) \tag{2}$$

where n is the number of bins representing PDF for a region and Z_{MOD} and Z_{OBS} are the count of values in any particular bin for the model and observed data, respectively. Any model with a Perkins score close to 1 means that the model data has a better PDF overlap with the observed data.

The precipitation score proposed by Sanchez is a comparison of model data with observational CDFs through five different factors. For the final skill score of precipitation, an equally weighted score from the Sanchez's and Taylor's scores is proposed (Sánchez et al. 2009; Taylor et al. 2012) and is presented in Eqs. 3a–3e:

$$f_1 = 1 - \left(\frac{\left|A_{\text{GCM}-} - A_{\text{PRISM}}\right|}{2.A_{\text{PRISM}}}\right)^{0.5} \tag{3a}$$

$$f_2 = 1 - \left(\frac{\left|A^+_{\text{GCM}} - A^+_{\text{PRISM}}\right|}{2.A^+_{\text{PRISM}}}\right)^{0.5} \tag{3b}$$

$$f_3 = 1 - \left(\frac{\left|A^-_{\text{GCM}} - A^-_{\text{PRISM}}\right|}{2.A^-_{\text{PRISM}}}\right)^{0.5} \tag{3c}$$

$$f_4 = 1 - \left(\frac{\overline{P_{\text{GCM}}} - \overline{P_{\text{PRISM}}}}{2.\overline{P_{\text{PRISM}}}}\right)^{0.5} \tag{3d}$$

$$f_5 = 1 - \left(\frac{\overline{\sigma_{\mathrm{GCM}}} - \overline{\sigma_{\mathrm{PRISM}}}}{2.\overline{\sigma_{\mathrm{PRISM}}}} \right)^{0.5} \quad (3e)$$

where A_{GCM} and A_{PRISM} are the areas below the cumulative distribution functions of the GCMs and PRISM, respectively; $A+$ and $A-$ are the areas right and left of the 50th percentile; *Overbar* denotes the spatial and time average; P is the mean annual precipitation; and σ is the standard deviation of the PDF. All of the five factors are not necessarily independent but represent various features of PDF of precipitation. The product of the five scores yields Sanchez's skill score for precipitation. The temperature and precipitation skill score are an equally weighted average of Taylor's scores and Perkins' or Sanchez's scores. The final skill score was computed by averaging precipitation and temperature skill score.

3.2 Downscaling and bias correction with quantile mapping

GCMs are global climate models with large horizontal resolutions ranging between 250 and 600 km. The GCM models are input to other hydrological models but suffer systematic errors in climate models due to unrealistic large-scale variability and unpredictable internal variability attributed to some physical processes. Hence, to avoid such issues and bridge the spatial scale issues, a systematic downscaling and bias correction approach must correct and minimize these biases. In this study, such correction was done here with reference to the high resolution 4 km PRISM precipitation and temperature dataset considered as a historical dataset (Daly and Bryant 2013).

Bias correction is a commonly used technique for climate change impacts studies, but it is not without flaws or limitations. These approaches employ corrections at various moments of the distribution, which can be mean, variance, skewness, or others depending on the meteorological variable or nature of the study. Bias corrections to all quantile of distribution are popular these days as it applies a correction to the full distribution, and it has outperformed the conventional bias correction schemes (Chen et al. 2013; Gudmundsson et al. 2012; Teutschbein and Seibert 2012, 2013). These are also better than parametric schemes as they are data-dependent, and most of the time, the lack of correct and long-term data proves a hindrance for parametric approaches.

However, one of the major flaws of the quantile mapping technique (QM) is a perturbation of the GCM projections (Maurer and Pierce 2014; Themeßl et al. 2012). The main hurdle arises from the assumption in QM that error correction functions among simulated and observed distributions show stationarity. It has been observed that due to the approach to preserve the quantiles, a perturbation in projected trends in seasonal mean precipitation encountered equivalent to original GCM-projected changes. Therefore, the quantile mapping approaches enhance the discrepancy, particularly when lesser skilled RCMs are used (Maraun 2013; Maurer and Pierce 2014; Themeßl et al. 2012). One of the main requirements is to preserve GCM signals such as trends to retain the model's climate sensitivity (Hempel et al. 2013). The detrended quantile mapping was developed to avoid this shortcoming, but this approach is not robust enough to preserve the more important trends in precipitation extremes (Bürger et al. 2012). The requirement is not to lose trends in all quantiles, and tails for extremes, supported by a modified version of this approach called quantile delta mapping (Cannon et al. 2015; Sunyer et al. 2014; Willems and Vrac 2011).

In the QM method, any modeled future value is bias-corrected with the same value of factor computed from historical and observed quantiles due to the stationarity approach. When applying QDM to any future precipitation value, QDM queries the observed values at the same percentile and multiplies it with the ratio of the future to historical modeled values at that particular quantiles. In contrast, for temperature, QDM adds the difference of the future and historical modeled values at the same quantiles to observed values. This method does not suffer from the stationarity and retains future projected signals. We used this method as a bias correction tool following the nearest neighborhood method that downscaled climate model scales to finer grids.

4 Results and discussion

4.1 GCM model selection

4.1.1 Selection based on changes in climatic extremes

The first step was focused on selecting the model that represents the change in climatic means over the ensemble pool in RCP4.5 and RCP8.5. In the first steps, model selection was done to represent the change in mean air temperature and annual precipitation between 2071–2100 and 1981–2005. First, four reference points were located. They represented four climate scenarios from a combination of wet and dry scenarios from precipitation changes with a warm and dry scenario from temperature changes. To make the process representative and avoid outliers, reference points are chosen at 10th and 90th percentile values, and models with the least distance from these points are the representative GCM models (Fig. 3). During this, 20 models per RCP were subset which would be refined in the further steps (Table 1).

The variation in climatic means for temperature varied from 0.1 to 3.9 °C in RCP4.5 and 2.3 to 6.4 °C in RCP8.5,

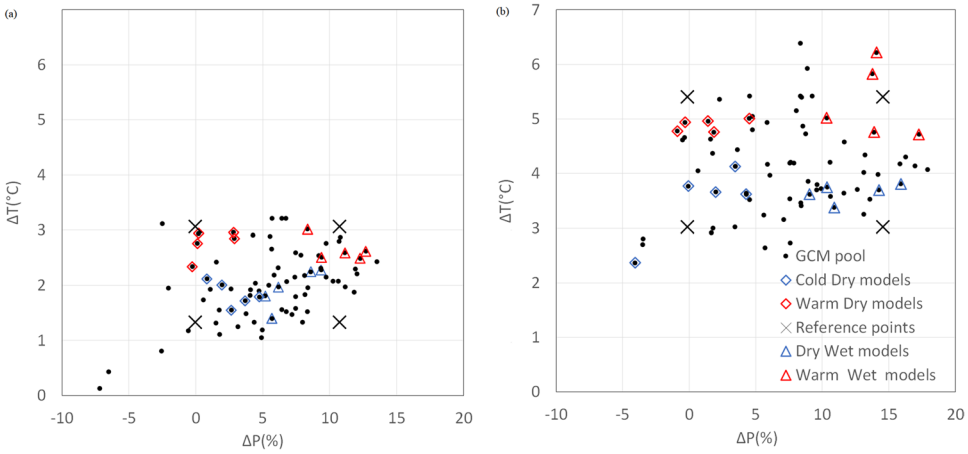


Fig. 3 Changes in climatic means in mean air temperature (ΔT) and annual precipitation sum (ΔP) between 1981–2005 and 2071–2100 for RCP4.5 (**a**) and RCP8.5 (**b**) GCM pool

and the variation in climate mean of annual precipitation varied from − 7.2 to 13.5% and − 4.0 to 20.1% in RCP8.5. The spread of the model in RCP8.5 and RCP4.5 was quite large, but the variability in the RCP8.5 scenario was higher and larger than in RCP4.5. The clustering of the model around the reference points in both RCP was not uniform as the degree of spread in temperature and precipitation was quite significant. One of the notable features, however, was the persistent clustering toward the corners at warmer conditions. The selected models drifted away from the reference points for drier conditions showing a great deal of variability.

4.1.2 Selection based on changes in climatic extremes

The second step was focused on selecting the models with extreme climate from the subset model pool derived the first step. Here, the four ETCCDI indices were used to define the extreme climate scenario. For example, the selected models were first ranked based on their ETCCDI indices representing extreme scenarios. A combined rank score representing precipitation and temperature extreme was derived to represent the model each of the four climatic scenarios, namely, cold and dry, wet and dry, cold and warm, and dry and warm (Table 2). For example, a combined rank score of ΔWSDI and ΔCDD was used to select the model for a warm and dry scenario, and ΔCSDI and ΔR95 were used to choose the model for cold and wet scenarios. This step was useful in better selection of climate models such that the spread

Table 2 Subset based on extreme indices for RCP4.5 and RCP8.5

RCP	Projections		CMIP Model	ΔWSDI (%)	ΔCDD (%)	ΔCSDI (%)	ΔR99 (%)	*P* score	*T* score	Final score
4.5	Dry	Cold	CSIRO-Mk3-6-0_r3i1p1	431	6.6	− 100.0	18.6	0.26	0.84	0.55
	Dry	Cold	CMCC-CM_r1i1p1	232	12.6	− 100.0	25.4	0.24	0.77	0.50
	Wet	Warm	CanESM2_r1i1p1	2113	17.6	− 100.0	20.5	0.47	0.77	0.62
	Wet	Warm	CanESM2_r5i1p1	2069	16.8	− 100.0	23.1	0.26	0.77	0.51
	Dry	Cold	CSIRO-Mk3-6-0_r9i1p1	1634	2.9	− 96.5	43.1	0.32	0.82	0.57
	Dry	Cold	GISS-E2-R_r6i1p3	652	7.1	− 83.2	50.9	0.30	0.86	0.58
	Wet	Warm	CSIRO-Mk3-6-0_r10i1p1	811	0.7	− 89.7	71.8	0.23	0.79	0.51
	Wet	Warm	CanESM2_r4i1p1	1669	6.7	− 97.8	52.9	0.54	0.78	0.66
8.5	Dry	Cold	CMCC-CM_r1i1p1	232	12.6	− 100.0	25.4	0.24	0.77	0.50
	Dry	Cold	IPSL-CM5A-MR_r1i1p1	− 8	53.1	− 10.5	21.6	0.33	0.79	0.56
	Wet	Warm	CanESM2_r1i1p1	2113	17.6	− 100.0	20.5	0.47	0.77	0.62
	Wet	Warm	HadGEM2-ES_r4i1p1	950	10.9	− 100.0	14.2	0.36	0.76	0.56
	Dry	Cold	CSIRO-Mk3-6-0_r7i1p1	724	0.4	− 100.0	46.5	0.19	0.82	0.51
	Dry	Cold	CSIRO-Mk3-6-0_r5i1p1	747	− 6.9	− 100.0	37.0	0.23	0.83	0.53
	Wet	Warm	HadGEM2-CC_r1i1p1	5.8	1.0	− 95.1	45.6	0.37	0.82	0.59
	Wet	Warm	CanESM2_r4i1p1	4.8	6.7	− 97.8	52.9	0.78	0.53	0.66

in a change in the climatic mean and climatic extreme. The general understanding was that models with larger changes in climatic means tend to have extreme climates as well, but this may not be necessarily true. Moreover, a combined scenario of the precipitation and temperature extreme index may exclude the model, which may be extreme in one of the index alone (Supplementary Fig. S1).

The first step helped in identifying 20 GCM models, which represent five climate models around each four-reference climatic scenario. In this step, the subset was reduced to 2 models per climatic scenario representing the extreme climate. The process is straightforward where the final index was the average of the precipitation and temperature climate indices. However, it may need careful selection when extreme indices were very near or equal to each other. For example, in a dry and cold scenario in RCP4.5, all the models lying in this category represented the extreme model from ΔCSDI values, but CMCC-CM_r1i1p1, CSIRO-Mk3-6-0_r3i1p1, and BCC-CSM1-1_r1i1p1 represent extreme model based on ΔCDD values. The combined rank score shows that BCC-CSM1-1_r1i1p1 and CSIRO-Mk3-6-0_r3i1p1 were two selected models from this step with a combined rank score of 4 and 3.5, respectively. BCC-CSM1-1_r1i1p1 had a similar score, but it lagged in precipitation index value compared to the values mentioned above. Another example is that, for a wet and cold scenario, GISS-E2-R_r6i1p3 had extreme values in both the indices selected for the first model; EC-EARTH_r8i1p, CSIRO-Mk3-6-0_r7i1p1, and CSIRO-Mk3-6-0_r9i1p10_r7i1p1 had equal final scores, making it challenging to choose from. The selection of the model from the rank score was difficult, so in this case, the ΔR95 and ΔCSDI were compared, and CSIRO-Mk3-6-0_r9i1p1 was selected as it has a median T index and P index among the other choices. There were no such issues in RCP8.5 due to the large spread of indices. Hence, the selection was based on the final combined score (Table 1).

4.1.3 Step 3: Selection based on model skill

The final step of the model selection was based on skill-based selection to find the final model subset for each possible scenario under each RCP scenario. The ensemble of the GCM pool showed a large degree of uncertainty and variability. The variation in annual climatic means for temperature varied from 0.1 to 3.2 °C in RCP4.5 and 2.4 to 6.4 °C in RCP8.5, and the variation in climate mean of annual precipitation varied from − 7.2 to 13.5% and − 4.0 to 17.9% in RCP8.5. Regarding the extreme climate, all GCM runs included in the selected ensembles project increased in warm spells and decreased in cold spells. Also, most of the models agreed to a significant increase in R95 values, although the change in annual mean for precipitation maybe not comparatively large. On the other hand, even though most of the models showed a wetter future, the number of consecutive dry days was also increasing. This accounts for the increasing variability in the future.

The final selected model matrix (Table 3) limits the uncertainty during the model selection process. In the final selected model subset, the variation in annual climatic means for temperature varied 1.7 to 3.0 °C in RCP4.5 and 2.4 to 5.8 °C in RCP8.5, and the variation in climate mean of annual precipitation varied from 0.2 to 12.7% in RCP4.5 and − 4.1 to 15.9% in RCP8.5. The change in WSDI(%) varied from 431 to 2112.6% for RCP4.5 and − 8.1 to 2112.6% for RCP8.5, and change in CDD varied from − 6.6 to 17.6% for RCP4.5 and − 6.9 to 53% for RCP8.5. The change in R95 varied from 18.6 to 52.9% for RCP4.5 and 14.2 to 46.5% for RCP8.5.

4.2 Bias correction and downscaling

The objective of this study was to find an optimal subset of the climate model from the GCM pool and employ quantile delta mapping to estimate potential future climate projection at the local level. The meteorological variables such as maximum and minimum daily temperature and daily precipitation were chosen so that the downscaled product could be used to drive hydrological models for climate impact studies.

4.2.1 Maximum temperature

There was considerable variation in the maximum temperature across the four prominent scenarios. However, the amount of variability within the interannual range was of higher significance. For the RCP4.5 scenario, there was an increase in the mean temperature in all months. This change was large for spring (MAM) and winter months (DJF), and also, the variability was very high with reference to the mean. As for the summer months (JJA), the monthly mean increase was small but quite apparent, and the variability was very less compared to other months. It is evident that even the cold scenarios had higher monthly means than the observed standard, and the warm scenarios have larger means and lesser variability compared to the cold scenarios. As for RCP8.5, the increase in monthly means was larger than all the months in RCP4.5.

Moreover, there was a higher perturbation of signals, which was significant for warmer scenarios than the cold scenarios. The increase in monthly and other percentiles was notable for the spring and winter months, and the summer months did not see much increment compared to the other months. The number of outliers and more extremes was more comparative in the RCP8.5 scenario.

The variability could also be attributed to the spatiotemporal averaging of the maximum temperature across all the

Table 3 Final subset models for RCP4.5 and RCP8.5

RCP	Projections		CMIP model	ΔP (%)	ΔT (°C)	ΔWSDI (%)	ΔCDD (%)	ΔCSDI (%)	ΔR95 (%)	*P* score	*T* score	Final scoring
4.5	Dry	Cold	CSIRO-Mk3-6-0_r3i1p1	3.7	1.7	431.0	6.6	− 100.0	18.6	0.26	0.84	0.55
	Dry	Warm	CanESM2_r1i1p1	0.2	2.9	2112.6	17.6	− 100.0	20.5	0.47	0.77	0.62
	Wet	Cold	GISS-E2-R_r6i1p3	5.2	1.8	652.0	7.1	− 83.2	50.9	0.30	0.86	0.58
	Wet	Warm	CanESM2_r4i1p1	8.3	3.0	1668.6	6.7	− 97.8	52.9	0.54	0.78	0.66
8.5	Dry	Cold	IPSL-CM5A-MR_r1i1p1	1.8	4.4	− 8.1	53.1	− 10.5	21.6	0.33	0.79	0.56
	Dry	Warm	HadGEM2-ES_r4i1p1	4.5	5.0	2121.1	10.9	− 100.0	14.2	0.36	0.76	0.56
	Wet	Cold	CSIRO-Mk3-6-0_r5i1p1	15.9	3.8	747.4	− 6.9	− 100.0	37.0	0.23	0.83	0.53
	Wet	Warm	CanESM2_r4i1p1	13.9	4.8	1668.6	6.7	− 97.8	52.9	0.78	0.53	0.66

grids of Rhode Island (Supplementary Figs. S2–S5). Rhode Island's coastal part had a milder change in increased mean maximum temperature than inland. The increment in change varied linearly from coastal to inland areas (Fig. 4).

4.2.2 Minimum temperature

There is evidence of more increased variation in the minimum temperature across the four prominent scenarios and in both RCP4.5 and 8.5. However, variability within the RCP8.5 was the largest for the minimum temperature. There was an increase in the mean temperature all month. However, the increment was largest for the winter. The increment in spring and fall (SON) looked similar, and the increment in the summer month seemed the least for RCP4.5 For RCP8.5, the variability of change was not uniform. The change in cold-warm and wet-warm scenarios was much higher than other similar two dry scenarios. The change in increased monthly mean for (JFMAM) first half of the year was highest for the cold-warm scenario, whereas for wet-warm, the fall months and first winter months had very high monthly means with increased variability. Again, for the summer months, the increase in monthly mean and the increased variability was small in comparison.

It can be observed that the change in monthly means and other percentiles was very large for minimum temperature compared to the maximum temperature. However, the difference in monthly means and other central tendencies was comparable for the summer months. The number of outliers and extremes was more evident in the RCP8.5 scenarios comparatively. The inherent spatial variability was not accounted for due to the spatiotemporal averaging (Supplementary Figs. S6–S9). The coastal part of Rhode Island had a milder change in increased mean minimum temperature than inland. Also, this increment in change varied more or less linearly from coastal to inland areas (Fig. 5).

4.2.3 Precipitation

The change in precipitation was not as prominent as temperature. Although studies have shown that in RCP4.5 the dry scenarios had lesser precipitation than observed data on an annual scale, large variability in the monthly scale and the winter month showed some increase in monthly means, but there was no explicit change in other months in the dry scenario in a monthly mean. In wet scenarios, the fall and winter months tended to have a higher change in monthly means with increased variability. In the case of RCP8.5, the change in monthly means was explicit. In general, all the months had increased means, but the change was significant in winter and spring. The summer months had very less changes in monthly mean and interannual variability. Similarly, the

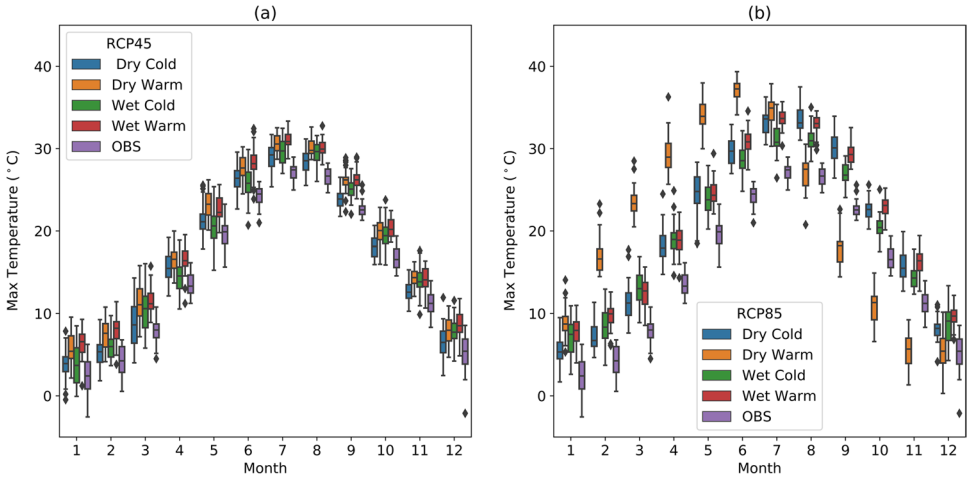


Fig. 4 Monthly variation in downscaled maximum temperature for 2071–2100 vs. observed gridded data in RCP4.5 (**a**) and RCP8.5 (**b**) GCM models

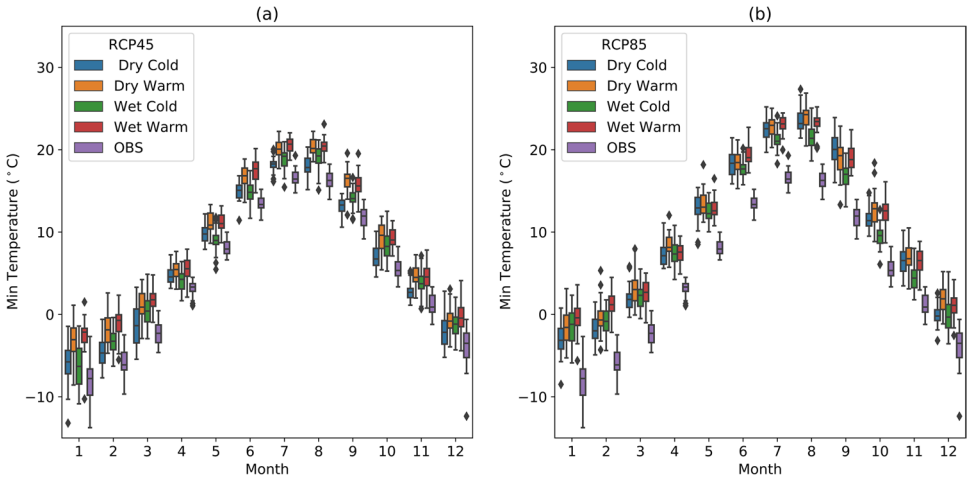


Fig. 5 Monthly variation in downscaled minimum temperature for 2071–2100 vs. observed gridded data in RCP4.5 (**a**) and RCP8.5 (**b**) GCM models

months with increased mean had high variability, and the presence of extreme events could also be noted.

The quantitative changes in precipitation in spatiotemporal domain were weak compared to temperature. This is consistent with studies where it has been argued that this may be attributed to the use of gridded data and the lack of statistically significant trend in observed precipitation (Agel et al. 2020; Rupp et al. 2013). However, along with the temporal variation, qualitative capturing of spatial variation in the monthly mean changes had been observed (Supplementary Figs. S10–S13). For the RCP4.5 scenario, there was little evidence of spatial variation in the dry scenario in winter, but for wet conditions, the inland areas were wetter than the coastal areas. For spring, in wet scenarios, all the months showed increased monthly means almost everywhere, but in dry scenarios, inland areas showed increased monthly means than coastal areas. This was also true for the summer and fall months except for some summer months showing decreased monthly means in summer and fall months, but overall, there was an increase in monthly means. For RCP8.5, although the change in monthly mean and median was strong, there was high variability throughout the annual periods. Except for some months, the changes were stronger inland than in the coastal areas (Fig. 6).

5 Conclusions

The approach of the study was to find an optimal subset of GCMs by addressing the inherent uncertainty and variability in the GCM pool. The selection process was robust as it gains strengths of envelope-based and skill-based methods and bridged the shortcomings of previous methods in model skill comparisons. However, it can be argued about that the order of the selection between the envelope and skill-based approach may lead to different models. There has been studies where skill-based selection leads the selection process, but the approach here was to include all possible GCM projections and retain the uncertainty and variability in the GCM matrix and apply bias correction techniques to tackle the issues of biases of GCMs with observation data.

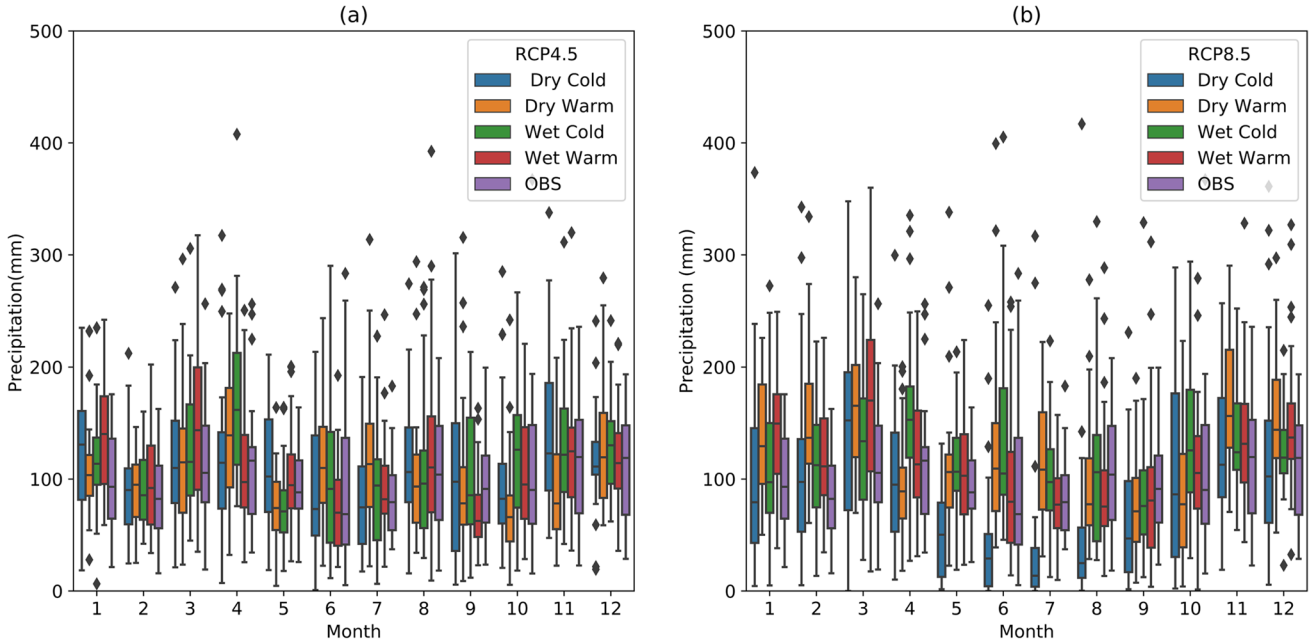


Fig. 6 Monthly variation in downscaled precipitation for 2071–2100 vs. observed gridded data in RCP4.5 (**a**) and RCP8.5 (**b**) GCM models

For skill-based selection, metrics such Taylor's score, Perkins' score, and Sanchez's score have been utilized to quantify model skills. A number of other skill metrics could have been applied as per the study objectives. The objective here was to address the necessity of both correlation and distribution-based skills at the same time.

The model selection started with all the GCM models available and its multiple ensemble members, which assumed that all the models were independent. Several studies have shown the evidence of sharing of model codes, forcing, and validation data, which violates the assumption, made (Knutti et al. 2013; Pennell and Reichler 2011; Pincus et al. 2008). However, the first step of subsetting through the envelope approach used here encourages picking the models from all plausible future projections.

It is a common practice to use GCM for future climate impact studies. Even though many GCMs lack the skill to simulate important regional climatic features, downscaling and bias correction methods are employed to reduce biases. Quantile delta mapping used here addresses the problem of stationarity, although this method is univariate. More robust approaches could be applied using other meteorological predictors with caution as required by study objectives.

Acknowledgements We acknowledge the World Climate Research Program's Working Group on Coupled Modeling for CMIP5 model output availability. We acknowledge Royal Netherlands Meteorological Institute for the KNMI Climate Explorer tool. In addition to that, we also acknowledge J. Sillmann for their output data on changes in climate change indices of CMIP5 models.

Author contribution SMP and SGS both conceived the presented idea. SGS performed the computations and analyzed the results and wrote the manuscript. SMP and SGS reviewed and edited the manuscript. SMP acquired funding to do this research.

Funding This work was supported by USDA RI0014- S1063, RI0021-S1089, McIntire Stennis RI0020-MS984, and Rhode Island HUD, Sandy grant for funding the research work.

Data availability All the output data used in this work are available at the University of Rhode Island and are free of charge.

Code availability Software used for the work is open source R: a Language and Environment for Statistical Computing.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Conflict of interest The authors declare no competing interests.

References

Agel L, Barlow M, Polonia J, Coe D (2020) Simulation of northeast US extreme precipitation and its associated circulation by CMIP5 models. J Clim 33:9817–9834

Anscombe FJ, 1973: Graphs in statistical analysis. *The american statistician*, **27,** 17–21 %@ 0003–1305.

Brekke LD, Dettinger MD, Maurer EP, Anderson M (2008) Significance of model credibility in estimating climate projection

distributions for regional hydroclimatological risk assessments. Clim Change 89:371–394
Bürger G, Murdock T, Werner A, Sobie S, Cannon A (2012) Downscaling extremes—an intercomparison of multiple statistical methods for present climate. J Clim 25:4366–4388
Cannon AJ, Sobie SR, Murdock TQ (2015) Bias correction of GCM precipitation by quantile mapping: how well do methods preserve changes in quantiles and extremes? J Clim 28:6938–6959
Change IC, 2013: The physical science basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. 2013. *There is no corresponding record for this reference.[Google Scholar]***,** 33–118.
Chen J, Brissette FP, Chaumont D, Braun M (2013) Performance and uncertainty evaluation of empirical downscaling methods in quantifying the climate change impacts on hydrology over two North American river basins. J Hydrol 479:200–214
Daly C, Bryant K (2013) The PRISM climate and weather system—an introduction. PRISM climate group, Corvallis
Eden JM, Widmann M, Maraun D, Vrac M (2014) Comparison of GCM- and RCM-simulated precipitation following stochastic postprocessing. J Geophys Res Atmos 119(11):040–011, 053
Evans JP, Ji F, Abramowitz G, Ekström M (2013) Optimally choosing small ensemble members to produce robust climate simulations. Environ Res Lett 8:044050
Flato, G., and Coauthors, 2014: Evaluation of climate models. *Climate change 2013: the physical science basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, Cambridge University Press, 741–866.
Friedlingstein P, Meinshausen M, Arora VK, Jones CD, Anav A, Liddicoat SK, Knutti R (2014) Uncertainties in CMIP5 climate projections due to carbon cycle feedbacks. J Clim 27:511–526
Gudmundsson L, Bremnes JB, Haugen JE, Engen-Skaugen T (2012) Downscaling RCM precipitation to the station scale using statistical transformations–a comparison of methods. Hydrol Earth Syst Sci 16:3383–3390
Gulizia C, Camilloni I (2015) Comparative analysis of the ability of a set of CMIP3 and CMIP5 global climate models to represent precipitation in South America. Int J Climatol 35:583–595
Hawkins E, Sutton R (2009) The potential to narrow uncertainty in regional climate predictions. Bull Am Meteor Soc 90:1095–1108
Hayhoe K, Coauthors (2006) Past and future changes in climate and hydrological indicators in the US Northeast. Clim Dyn 28:381–407
Hempel S, Frieler K, Warszawski L, Schewe J, Piontek F, 2013: A trend-preserving bias correction–the ISI-MIP approach.
Hidalgo HG, Alfaro EJ (2012) Global model selection for evaluation of climate change projections in the Eastern Tropical Pacific Seascape. Rev Biol Trop 60:67–81
Hidalgo HG, Alfaro EJ (2015) Skill of CMIP5 climate models in reproducing 20th century basic climate features in Central America. Int J Climatol 35:3397–3421
Hodgkins GA, Dudley RW (2006) Changes in late-winter snowpack depth, water equivalent, and density in Maine, 1926–2004. Hydrol Process 20:741–751
Hodgkins GA, Dudley RW, 2006b: Changes in the timing of winter–spring streamflows in eastern North America, 1913–2002. *Geophys Res Lett* **33**.
Hodgkins GA, James IC, Huntington TG (2002) Historical changes in lake ice-out dates as indicators of climate change in New England, 1850–2000. Int J Climatol 22:1819–1827
Hodgkins GA, Dudley RW, Huntington TG (2003) Changes in the timing of high river flows in New England over the 20th century. J Hydrol 278:244–252
Immerzeel W, Pellicciotti F, Bierkens M (2013) Rising river flows throughout the twenty-first century in two Himalayan glacierized watersheds. Nat Geosci 6:742–745
Karmalkar AV, Horton RM (2021) Drivers of exceptional coastal warming in the northeastern United States. Nat Clim Chang 11:854–860
Kjellström E, Boberg F, Castro M, Christensen JH, Nikulin G, Sánchez E (2010) Daily and monthly temperature and precipitation statistics as performance indicators for regional climate models. Clim Res 44:135–150
Knutti R, Sedláček J (2012) Robustness and uncertainties in the new CMIP5 climate model projections. Nat Clim Chang 3:369–373
Knutti R, Masson D, Gettelman A (2013) Climate model genealogy: generation CMIP5 and how we got there. Geophys Res Lett 40:1194–1199
Kunkel K, Coauthors, 2022: State climate summaries for the United States 2022. NOAA Technical Report NESDIS 150.
Liu J, Song M, Horton RM, Hu Y (2013) Reducing spread in climate model projections of a September ice-free Arctic. Proc Natl Acad Sci U S A 110:12571–12576
Lutz AF, ter Maat HW, Biemans H, Shrestha AB, Wester P, Immerzeel WW (2016) Selecting representative climate models for climate change impact studies: an advanced envelope-based selection approach. Int J Climatol 36:3988–4005
Maraun D (2013) Bias correction, quantile mapping, and downscaling: revisiting the inflation issue. J Clim 26:2137–2143
Maurer EP, Pierce DW (2014) Bias correction can modify climate model simulated precipitation changes without adverse effect on the ensemble mean. Hydrol Earth Syst Sci 18:915–925
McSweeney CF, Jones RG (2016) How representative is the spread of climate projections from the 5 CMIP5 GCMs used in ISI-MIP? Clim Serv 1:24–29
Pennell C, Reichler T (2011) On the effective number of climate models. J Clim 24:2358–2367
Perkins SE, Pitman AJ, Holbrook NJ, McAneney J (2007) Evaluation of the AR4 climate models' simulated daily maximum temperature, minimum temperature, and precipitation over Australia using probability density functions. J Clim 20:4356–4376
Peterson T (2005) Climate Change Indices. WMO Bull 54:83–86
Pierce DW, Barnett TP, Santer BD, Gleckler PJ (2009) Selecting global climate models for regional climate change studies. Proc Natl Acad Sci U S A 106:8441–8446
Pincus R, Batstone CP, Hofmann RJP, Taylor KE, Glecker PJ, 2008: Evaluating the present-day simulation of clouds, precipitation, and radiation in climate models. *J Geophys Re Atmos* **113**.
Rosen RA, Guenther E (2015) The economics of mitigating climate change: what can we know? Technol Forecast Soc Chang 91:93–106
Ruane AC, SP McDermid, 2017: Selection of a representative subset of global climate models that captures the profile of regional changes for integrated climate impacts assessment. *Earth Perspect* **4**.
Rupp DE, Abatzoglou JT, Hegewisch KC, Mote PW (2013) Evaluation of CMIP5 20th century climate simulations for the Pacific Northwest USA. J Geophys Res Atmos 118(10):884–810, 906
San José R, Pérez JL, González RM, Pecci J, Garzón A, Palacios M (2016) Impacts of the 4.5 and 8.5 RCP global climate scenarios on urban meteorology and air quality: application to Madrid, Antwerp, Milan, Helsinki and London. J Comput Appl Math 293:192–207
Sánchez E, Romera R, Gaertner MA, Gallardo C, Castro M (2009) A weighting proposal for an ensemble of regional climate models over Europe driven by 1961–2000 ERA40 based on monthly precipitation probability density functions. Atmos Sci Lett 10:241–248
Sillmann J, Kharin V, Zhang X, Zwiers F, Bronaugh D (2013a) Climate extremes indices in the CMIP5 multimodel ensemble: Part 1. Model evaluation in the present climate. J Geophys Res Atmos 118:1716–1733

Sillmann J, Kharin VV, Zwiers F, Zhang X, Bronaugh D (2013b) Climate extremes indices in the CMIP5 multimodel ensemble: Part 2. Future climate projections. J Geophys Res Atmos 118:2473–2493

Sorg A, Huss M, Rohrer M, Stoffel M (2014) The days of plenty might soon be over in glacierized Central Asian catchments. Environ Res Lett 9:104018

Sunyer MA, Coauthors (2014) Inter-comparison of statistical downscaling methods for projection of extreme precipitation in Europe. Hydrol Earth Syst Sci Discuss 11:6167–6214

Taylor KE (2001) Summarizing multiple aspects of model performance in a single diagram. J Geophys Res Atmos 106:7183–7192

Taylor KE, Stouffer RJ, Meehl GA (2012) An overview of CMIP5 and the experiment design. Bull Am Meteor Soc 93:485–498

Teutschbein C, Seibert J (2012) Bias correction of regional climate model simulations for hydrological climate-change impact studies: review and evaluation of different methods. J Hydrol 456:12–29

Teutschbein C, Seibert J (2013) Is bias correction of regional climate model (RCM) simulations possible for non-stationary conditions? Hydrol Earth Syst Sci 17:5061–5077

Themeßl MJ, Gobiet A, Heinrich G (2012) Empirical-statistical downscaling and error correction of regional climate models and its impact on the climate change signal. Clim Change 112:449–468

Tu J (2009) Combined impact of climate and land use changes on streamflow and water quality in eastern Massachusetts, USA. J Hydrol 379:268–283

Van Vuuren DP, Coauthors, 2011: The representative concentration pathways: an overview. *Clim Change*, **109(**5)

Warszawski L, Frieler K, Huber V, Piontek F, Serdeczny O, Schewe J (2014) The inter-sectoral impact model intercomparison project (ISI-MIP): project framework. Proc Natl Acad Sci U S A 111:3228–3232

Willems P, Vrac M (2011) Statistical precipitation downscaling for small-scale hydrological impact investigations of climate change. J Hydrol 402:193–205

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.